

Quasilinear Indistinguishability Obfuscation

via

Propositional Proofs of Equivalence

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Joint work with Yaohua Ma, Chenxin Dai



Indistinguishability Obfuscation

[GGHRSW'13]

**Candidate
iO**



[GGHRSW'13]

Candidate
iO



[JLS'20]

iO from
well-founded
assumptions



many attempts

Feasibility of provably secure iO?

[GGHRSW'13]

Candidate
iO



[JLS'20]

iO from
well-founded
assumptions



Efficiency of iO ?

Provably secure iO

 Polynomial blowup

 Input-length barrier?

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Provably secure iO

 Polynomial blowup

 Input-length barrier?

Heuristic iO

 Quasilinear efficiency

- VBB-obf for PRF $\Rightarrow$ full obf [Applebaum]
- VBB-obf + SNARG $\Rightarrow$ full obf [Boneh's talk]

Efficiency of iO ?

Provably secure iO

 Polynomial blowup

 Input-length barrier?

Heuristic iO

 Quasilinear efficiency

- VBB-obf for PRF $\Rightarrow$ full obf [Applebaum]
- VBB-obf + SNARG $\Rightarrow$ full obf [Boneh's talk]

Can we have the best of both worlds?

Provably secure iO

 Quasilinear efficiency

Our Result

Provably secure iO

 Quasilinear efficiency



via

Propositional proof of equivalence

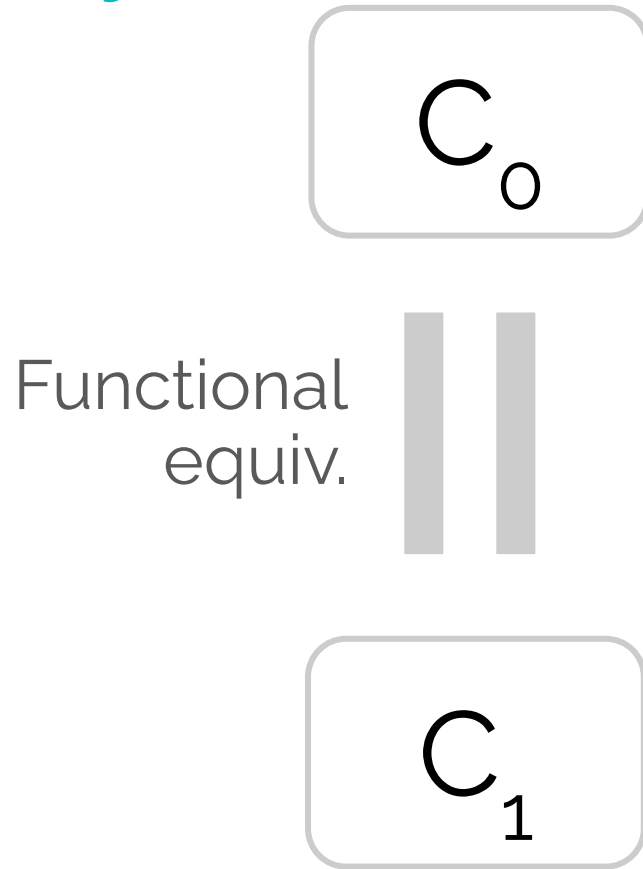
Our Result

What is

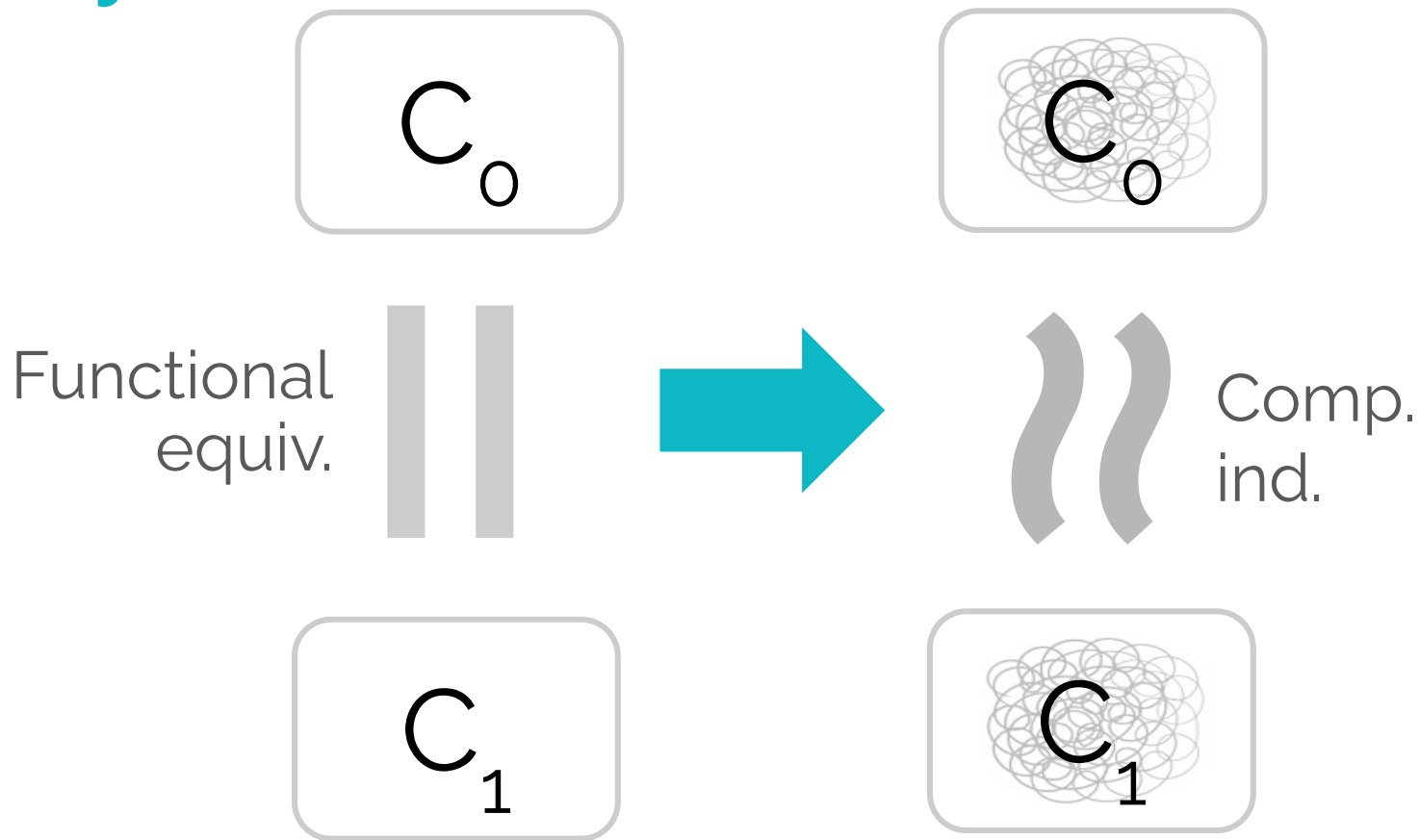
Propositional proof of equivalence

and why?

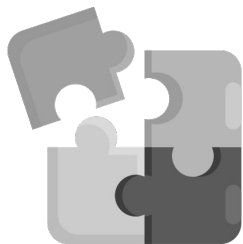
iO Security



iO Security

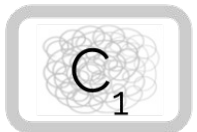
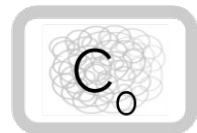


Hard problem



Reduction R

iO
adversary



Conjecture: R “knows” proof of equiv



Conjecture: R “knows” proof of equiv



Conjecture: R “knows” proof of equiv

$\Rightarrow$ R must enumerate all inputs



Conjecture: R “knows” proof of equiv

⇒ R must enumerate all inputs

⇒ # hybrids exponential in input len

Not a formal proof

Conjecture: R “knows” proof of equiv

⇒ R must enumerate all inputs

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⇒ $\text{sec param} \geq \text{poly}(\text{input len})$

Not a formal proof

Conjecture: R “knows” proof of equiv

⇒ R must enumerate all inputs

⇒ # hybrids exponential in input len

⇒ $\text{sec param} \geq \text{poly}(\text{input len})$

“
folklore:
The input length barrier”



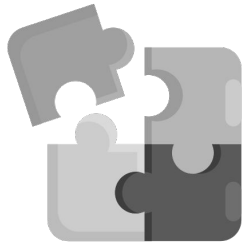
Overcoming the input-len barrier



Proof: $C_0 = C_1$

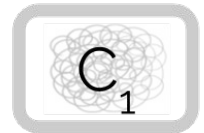
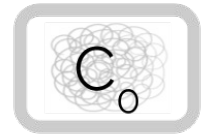


Hard problem



Reduction R

iO
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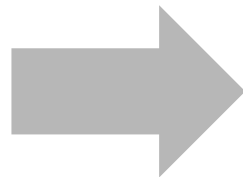
Relaxed iO security


Short proof:
 $C_0 = C_1$

C_0



C_1



C_0



C_1

Expressive power of relaxed notion ?

“
[

Jain and Jin:

Sufficient for almost all
known applications of iO

”
]



's blueprint



Short proof: $C = C^*$



Padded
 C

=

C_1

=

C_2

=

...

=

Padded
 C^*

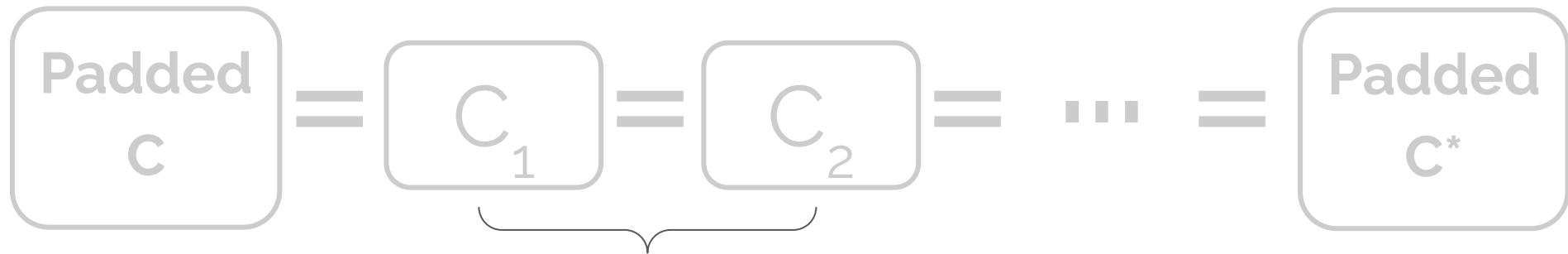
Identical except an $O(\log n)$ -sized subcircuit



's blueprint

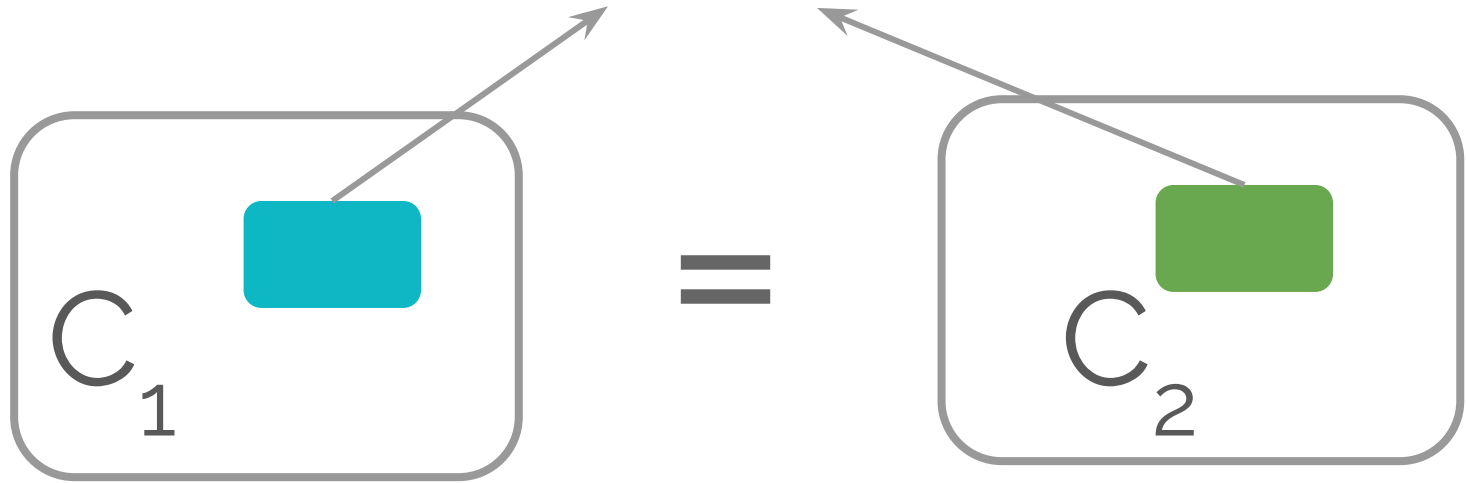


Short proof: $\mathbf{C} = \mathbf{C}^*$



$O(\log n)$ -equivalent

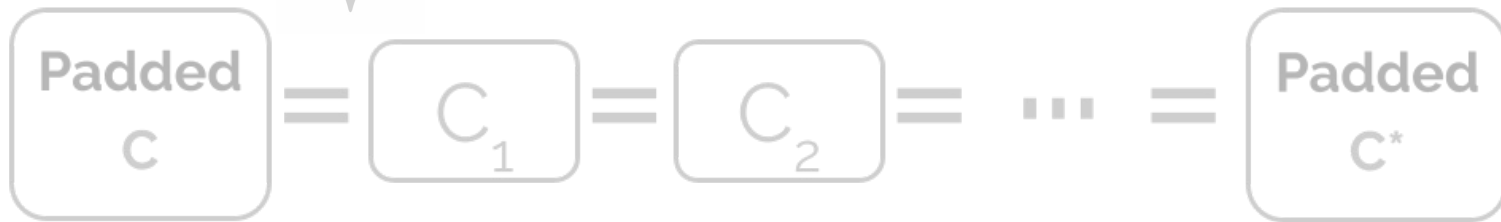
Functionally equiv, differing in impl
 $O(\log n)$ size



$O(\log n)$ -equivalent circuits



Short proof: $\mathbf{C} = \mathbf{C}^*$

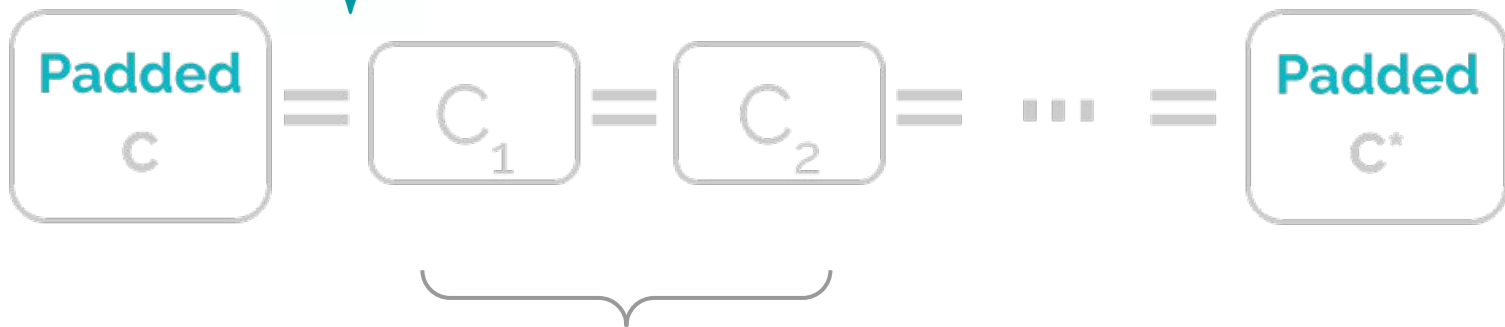


iO for $O(\log n)$ -equivalent circuits suffices

Challenge 1



Short proof: $\mathbf{C} = \mathbf{C}^*$



iO for $O(\log n)$ -equivalent circuits

Challenge 2



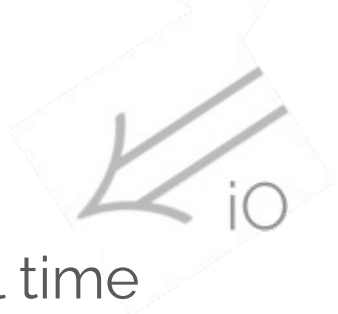
's approach

n gates



n^2 gates

$\geq n^4$ eval time



assume: BARG with
ideal efficiency (not known)

Challenge 1

padding

iO for log-equiv circ

Challenge 2

Our approach

n gates



$\sim n$ gates

$\sim n$ eval time

assume: proof size $\sim$ circuit size

Challenge 1

padding

iO for log-equiv circ

Challenge 2

Challenge 1

padding

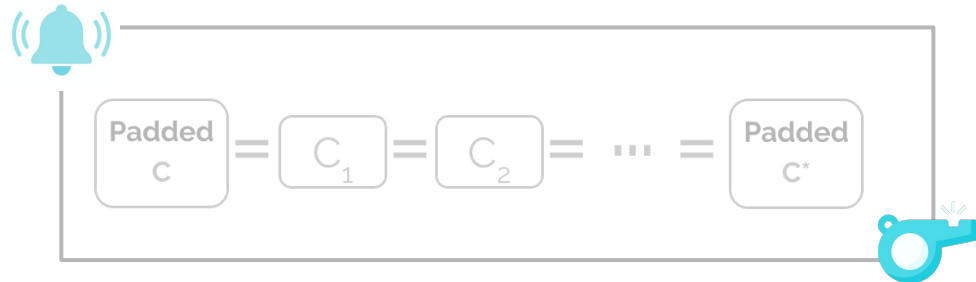
Challenge 1

padding

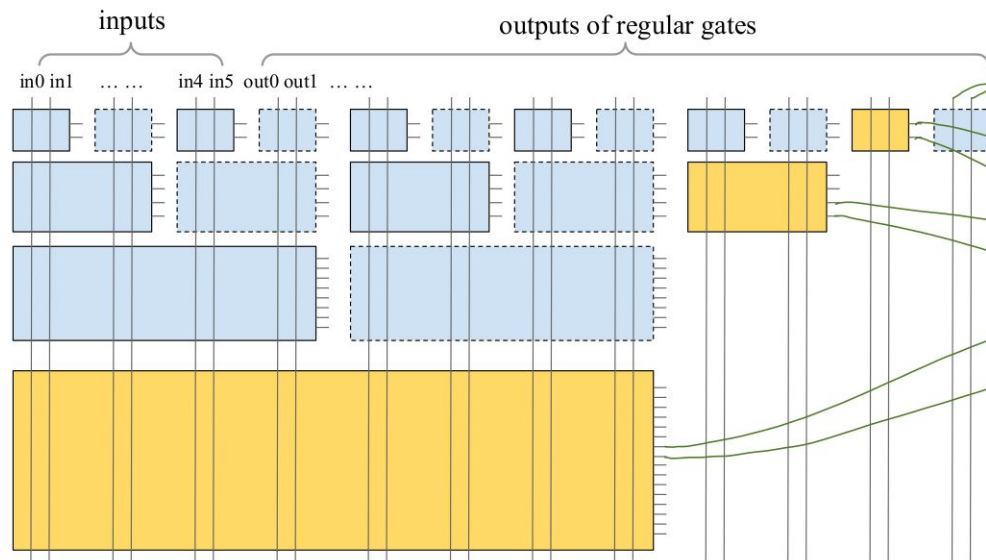


Padded circuit is a **universal circuit**

But with



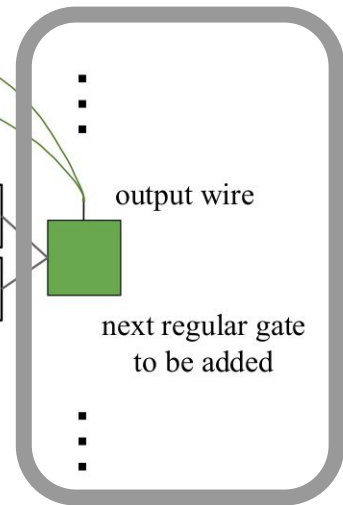
1 Routing gates



Routing network Not needed Currently active

Challenge 1

padding



2 Regular gates

Challenge 2

iO for log-equiv circ



Gate by gate obfuscation



Mix-and-match attack

e.g., use values from a different eval

Challenge 2

iO for log-equiv circ



each wire carries a BARG provenance proof, $O(n)$ time per wire

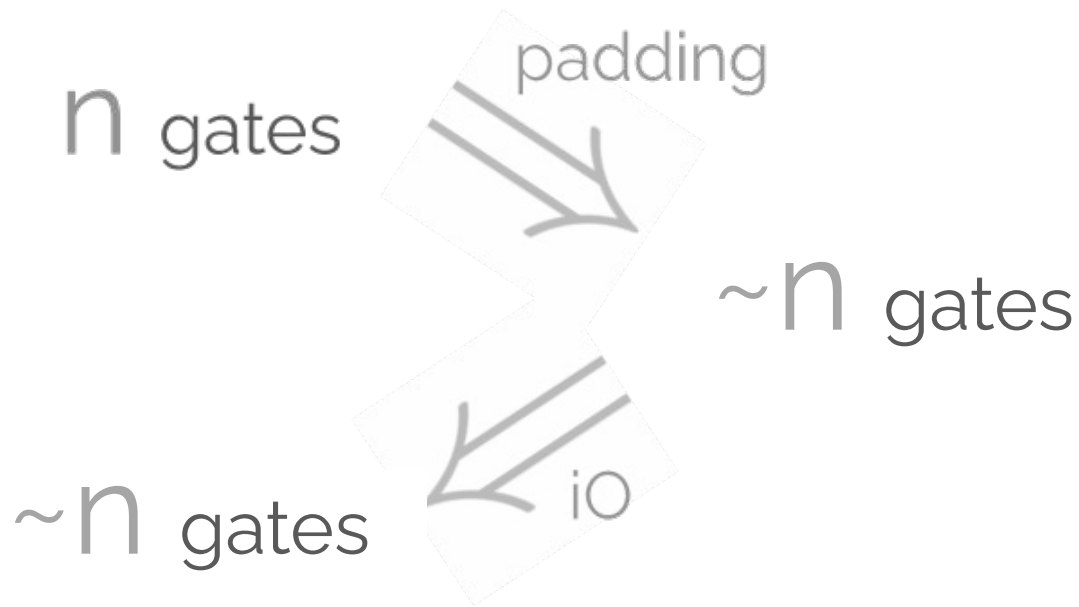
Ours

achieve the same but w/o BARG,
 $\sim O(1)$ time per wire

Applications of our quasilinear iO

- **Multi-input functional encryption** with quasilinear efficiency
- **iO for TM** with quasilinear efficiency

assume: short proofs of equivalence



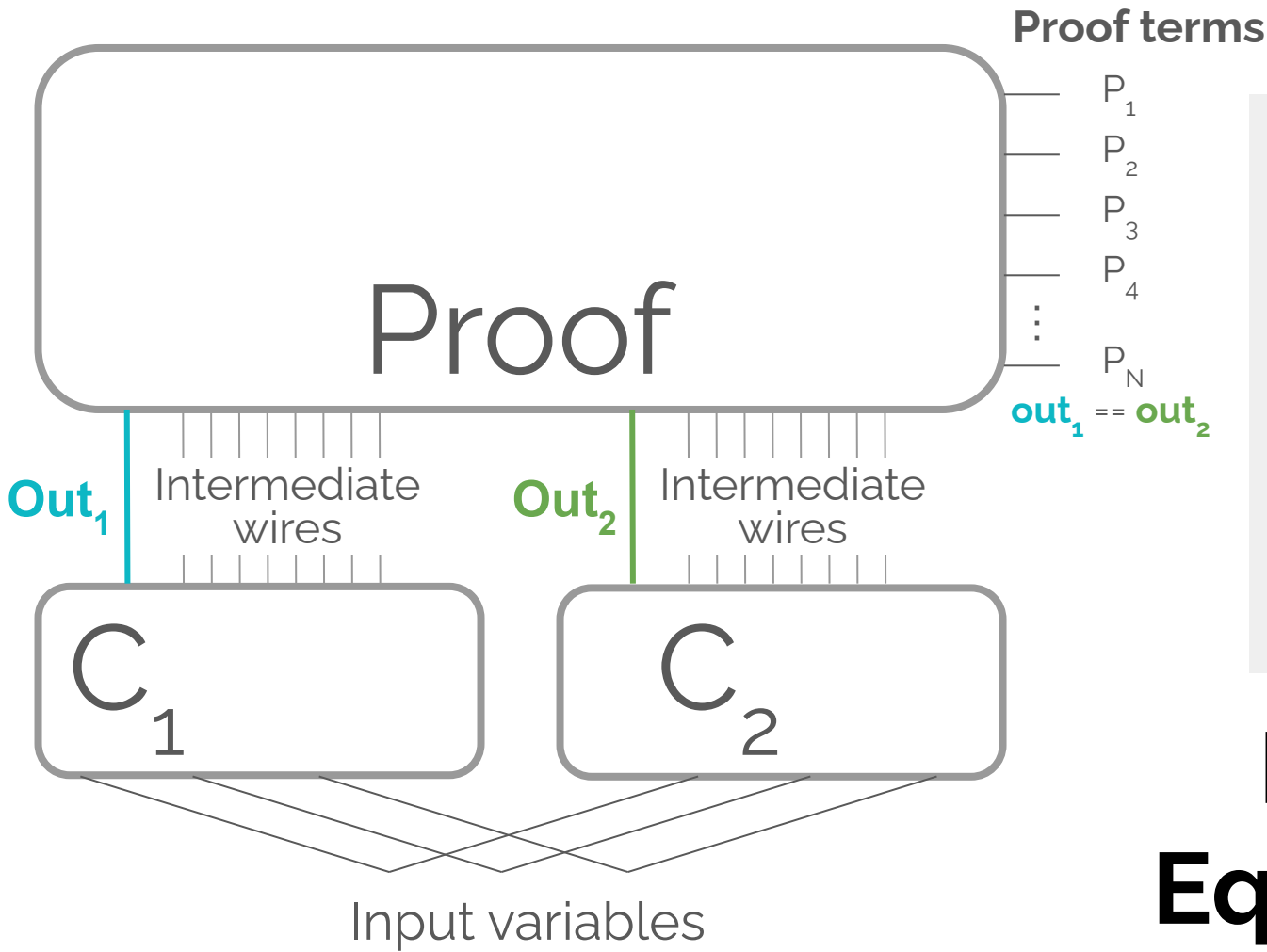
assume: proof size $\sim$ circuit size

Challenge 1

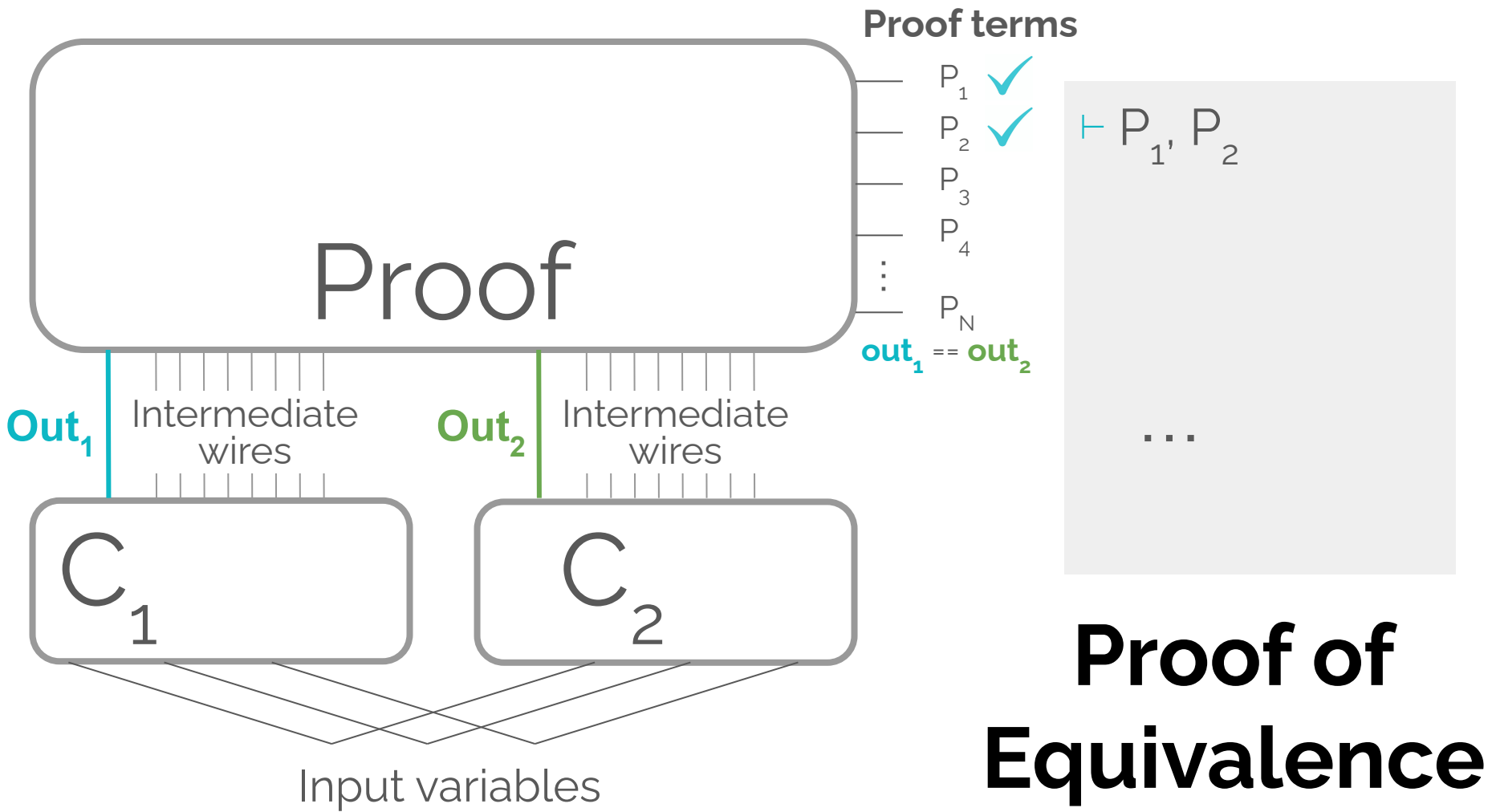
padding

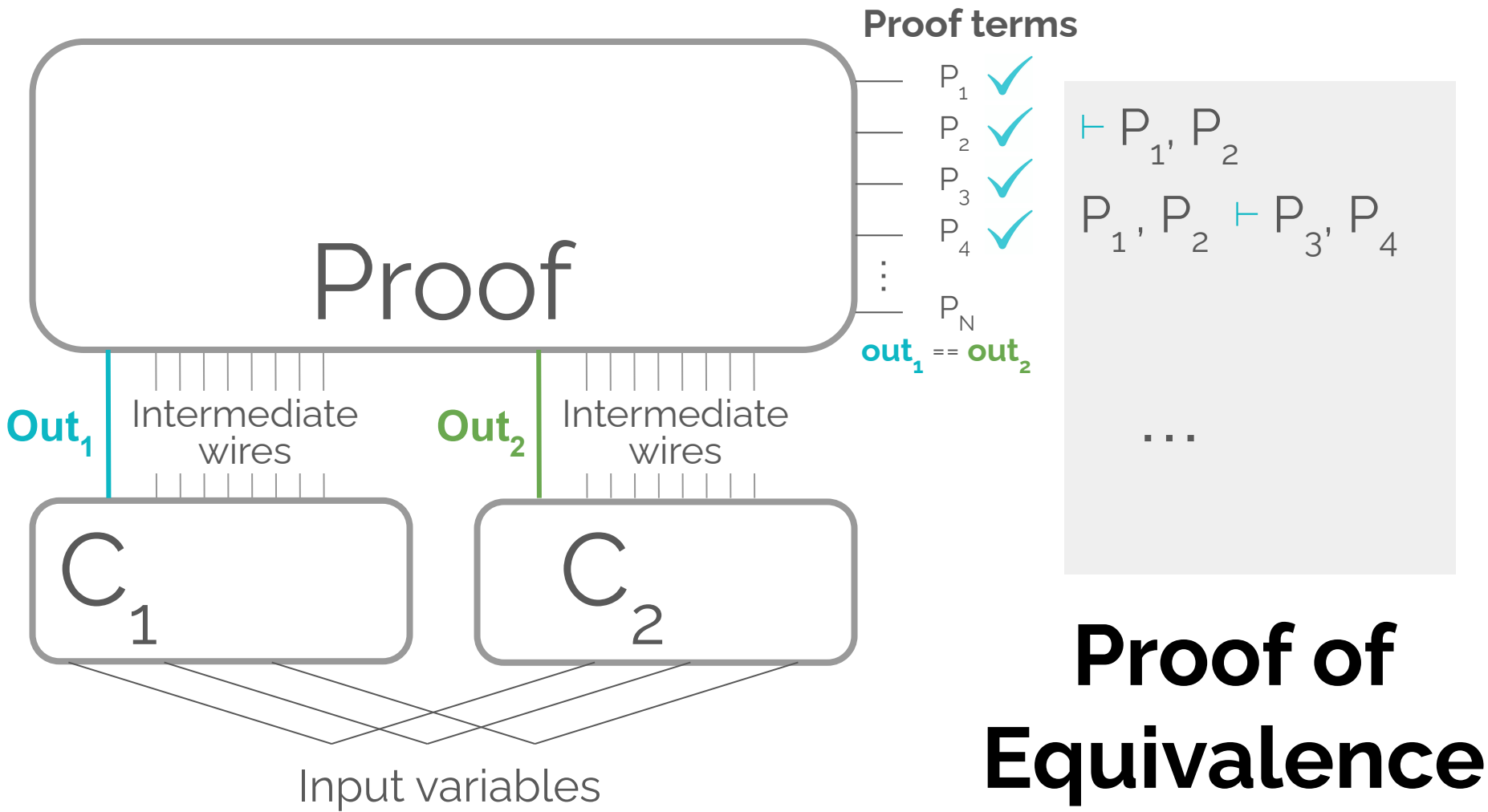
iO for log-equiv circ

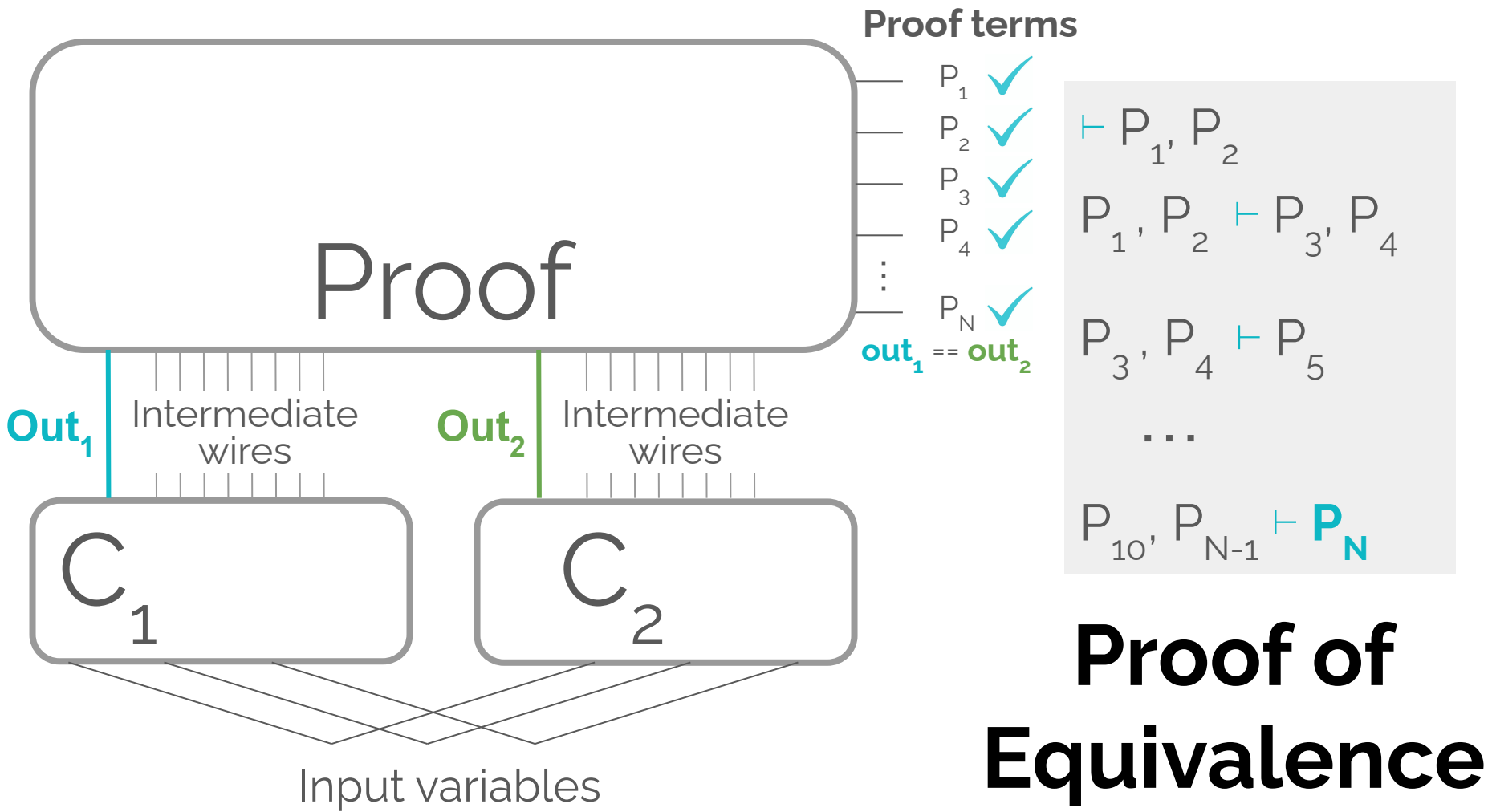
Challenge 2



Proof of Equivalence







Axioms { $\vdash P \rightarrow (Q \rightarrow P)$
 $\vdash \neg\neg P \rightarrow P$

...

Modus
Ponens $P, P \rightarrow Q \vdash Q$



$P_3, P_4 \vdash P_5$

Example of a Proof Line

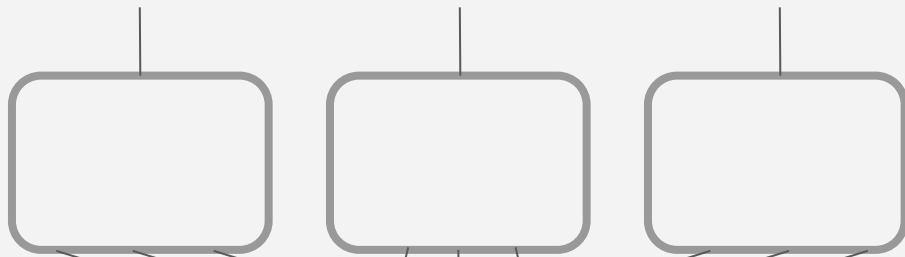
$O(1)$ subcircuit



P_3

P_4

P_5



Q_1

Q_2

Q_3

$O(1)$ variables

$P_3, P_4 \vdash P_5$

Each proof line: $O(1)$ -sized subcircuit

Example: $\nexists$ Short Equivalence Proof

$R \notin \text{img}(\text{PRG})$

// do something

If $\text{PRG}(\text{inp}) = R$ **then**

// do special

Else

// proceed normally

=

// do something

~~**If** $\text{PRG}(\text{inp}) = R$ **then**~~

~~// do special~~

~~**Else**~~

~~// proceed normally~~

[SW13]

Example: $\exists$ Short Equivalence Proof

C : encryption of 1

// do something

If $\text{Enc}(o, r) = C$ **then**

// do special

Else

// proceed normally

=

// do something

~~**If** $\text{Enc}(o, r) = C$ **then**~~

~~// do special~~

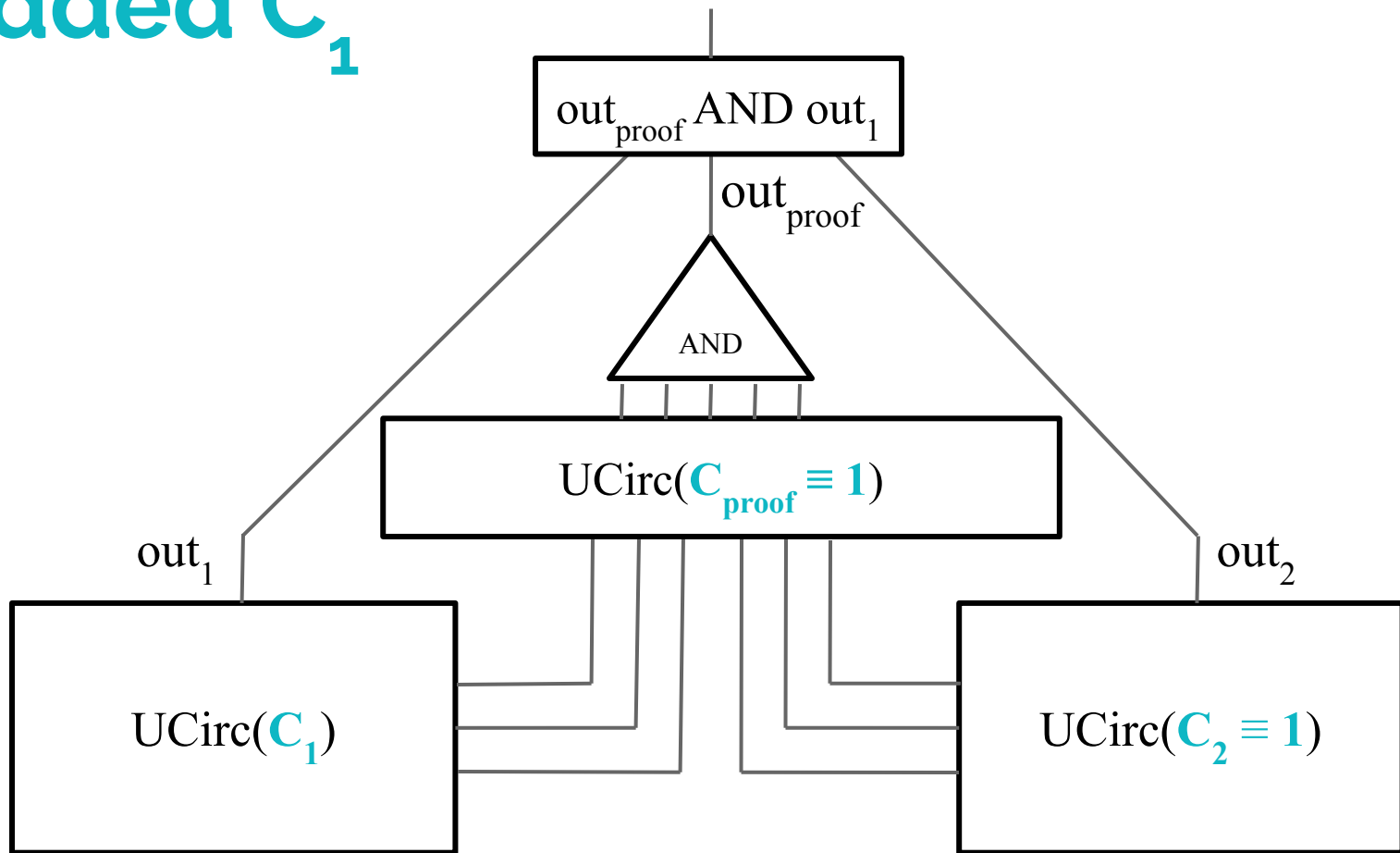
~~**Else**~~

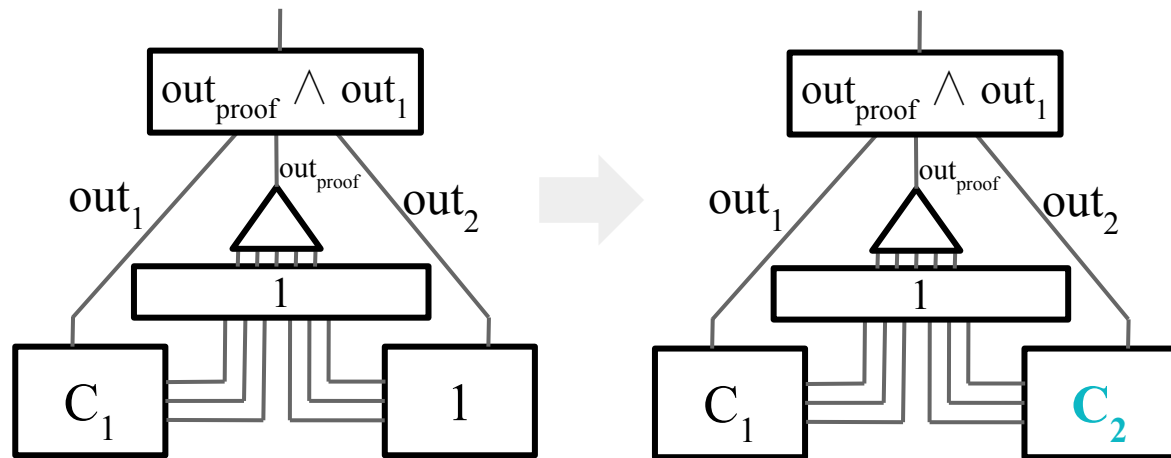
~~// proceed normally~~

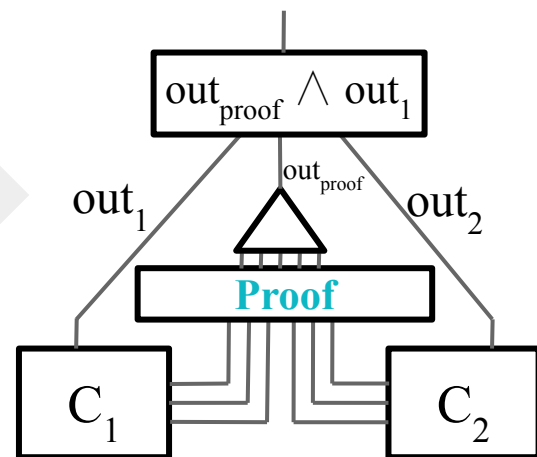
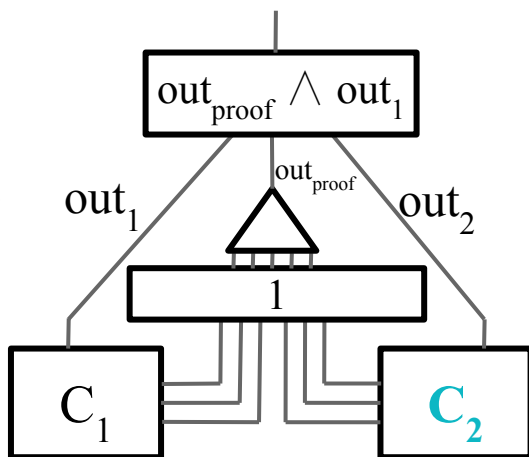
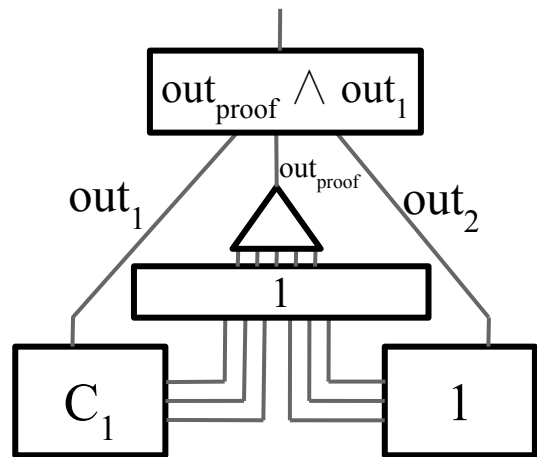


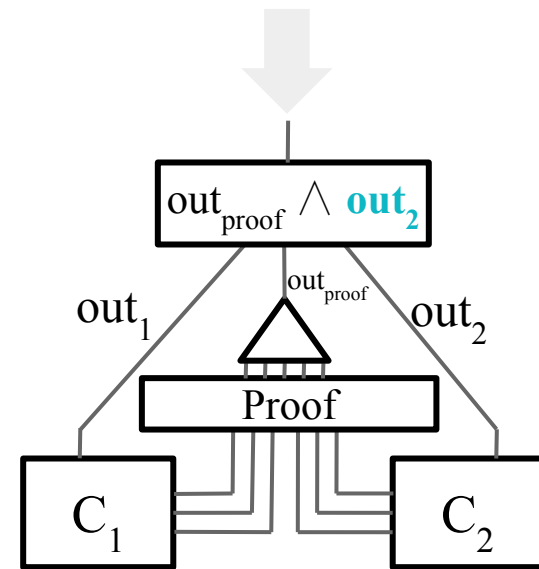
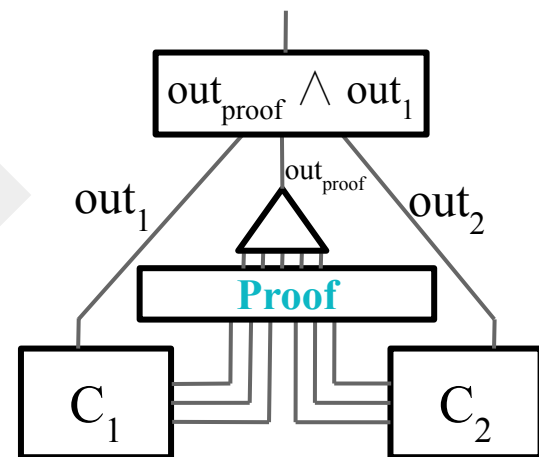
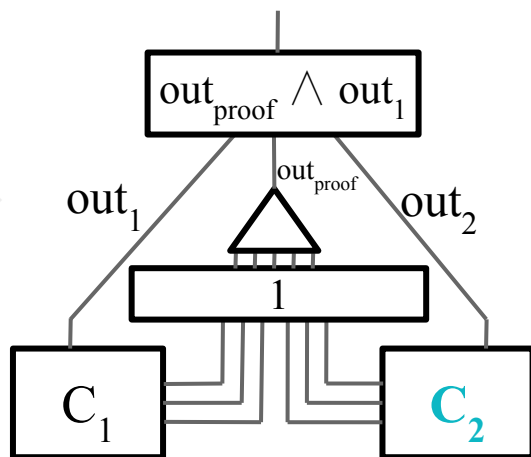
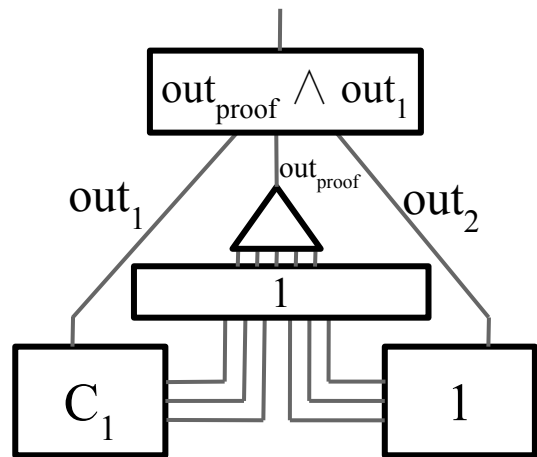
PKE w/ short proof of correct decryption

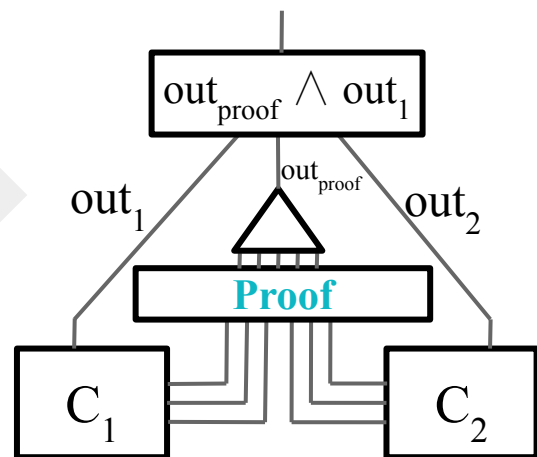
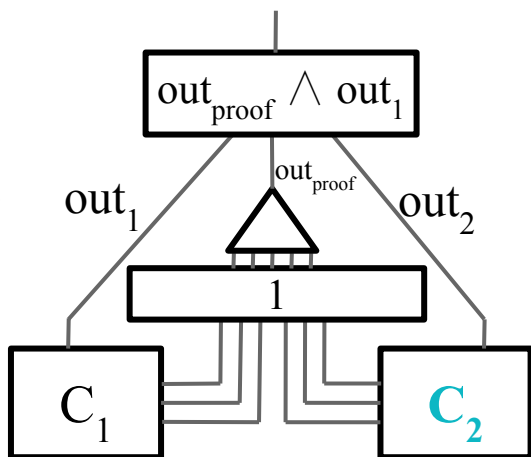
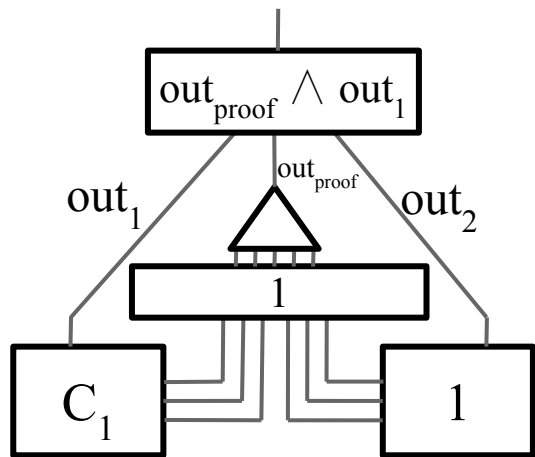
Padded C_1



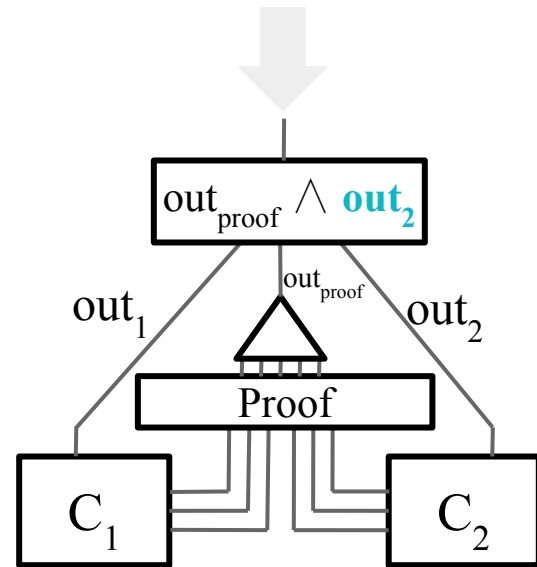


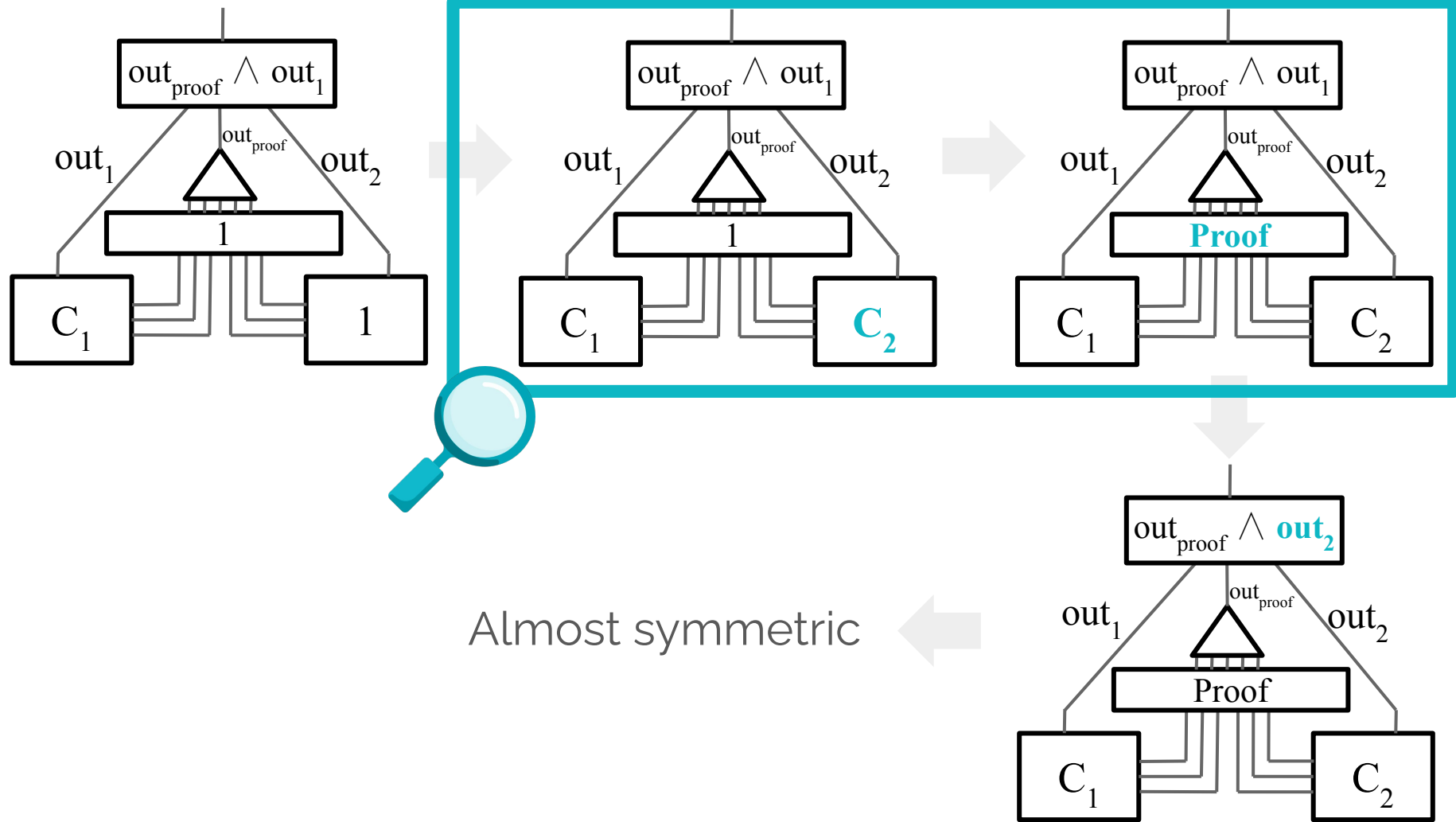




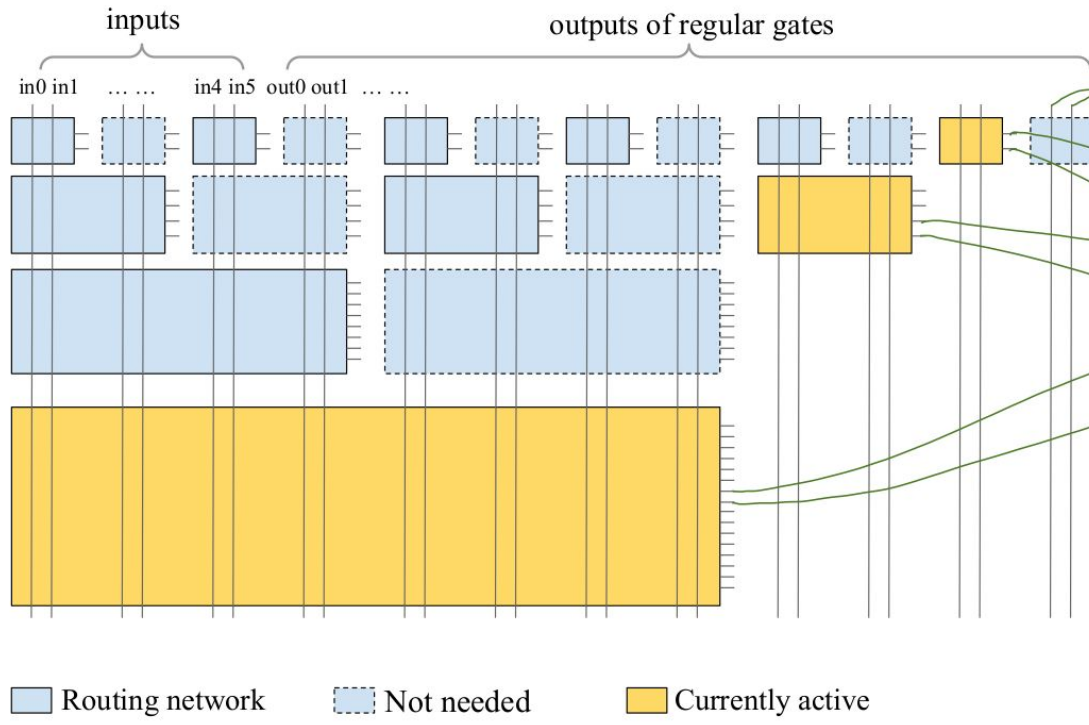


Almost symmetric

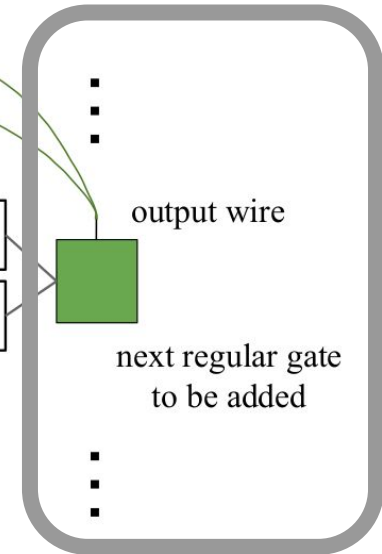




1 Routing gates

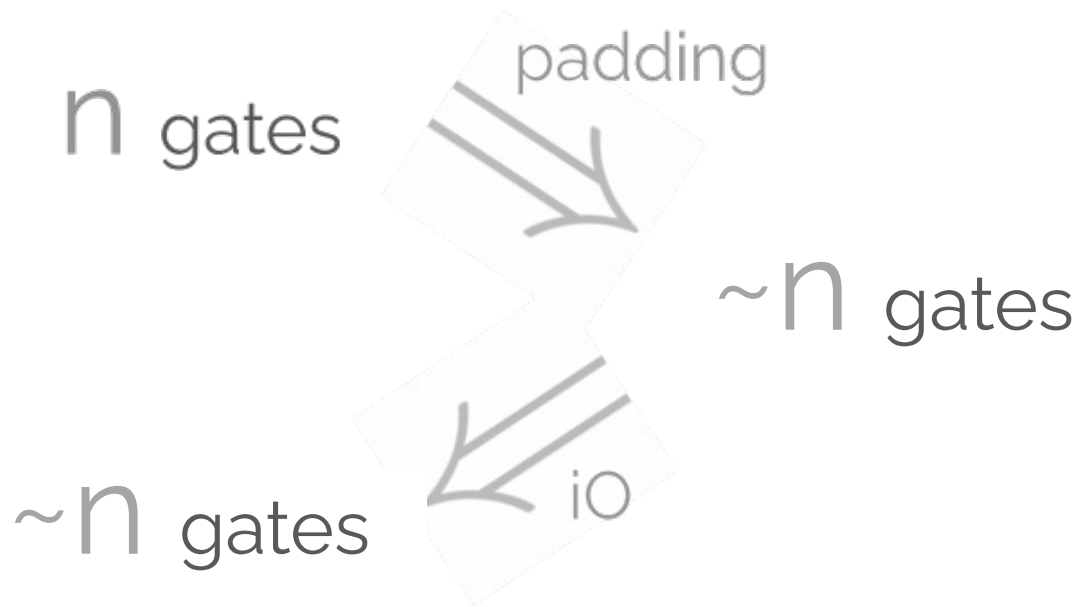


2 Regular gates



Challenge 1

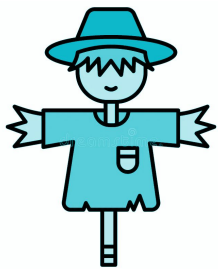
padding



assume: proof size $\sim$ circuit size

iO for log-equiv circ

Challenge 2



$ct_1, \sigma_1 \mid \mid ct_2, \sigma_2$

Assert σ_1, σ_2 are valid MACs for ct_1, ct_2 using K_{mac}^1, K_{mac}^2

$m_1, m_2 = \text{Dec}(ct_1, ct_2)$ using K_{enc}^1, K_{enc}^2

Compute $out = g(m_1, m_2)$

Output $ct' = \text{Enc}(out), \sigma' = \text{MAC}(ct')$ using $K_{enc}^{out}, K_{MAC}^{out}$



can mix ct, σ from 2 evaluations

ct_1, σ_1 | ct_2, σ_2

Assert σ_1, σ_2 are valid MACs for ct_1, ct_2 using K_{mac}^1, K_{mac}^2

$m_1, m_2 = \text{Dec}(ct_1, ct_2)$ using K_{enc}^1, K_{enc}^2

Compute $out = g(m_1, m_2)$

Output $ct' = \text{Enc}(out), \sigma' = \text{MAC}(ct')$ using $K_{enc}^{out}, K_{MAC}^{out}$

ct_1, h_1, σ_1

ct_2, h_2, σ_2

h' ,



π

Assert π proves that (ct_1, ct_2, h_1, h_2) consistent w/ h'

Assert σ_1, σ_2 valid for $(ct_1, h_1), (ct_2, h_2)$ using K_{mac}^1, K_{mac}^2

$m_1, m_2 = \text{Dec}(ct_1, ct_2)$ using K_{enc}^1, K_{enc}^2

Compute $out = g(m_1, m_2)$

Output $ct' = \text{Enc}(out), \sigma' = \text{MAC}(ct', h')$ using $K_{enc}^{out}, K_{MAC}^{out}$

ct_1, h_1, σ_1

ct_2, h_2, σ_2

h'



Assert π proves that (ct_1, ct_2, h_1, h_2) consistent w/ h'

SSB
hashes

BARG
proof

Compute $out = g(m_1, m_2)$

Output $ct' = Enc(out), \sigma' = MAC(ct', h')$ using $K_{enc}^{out}, K_{MAC}^{out}$

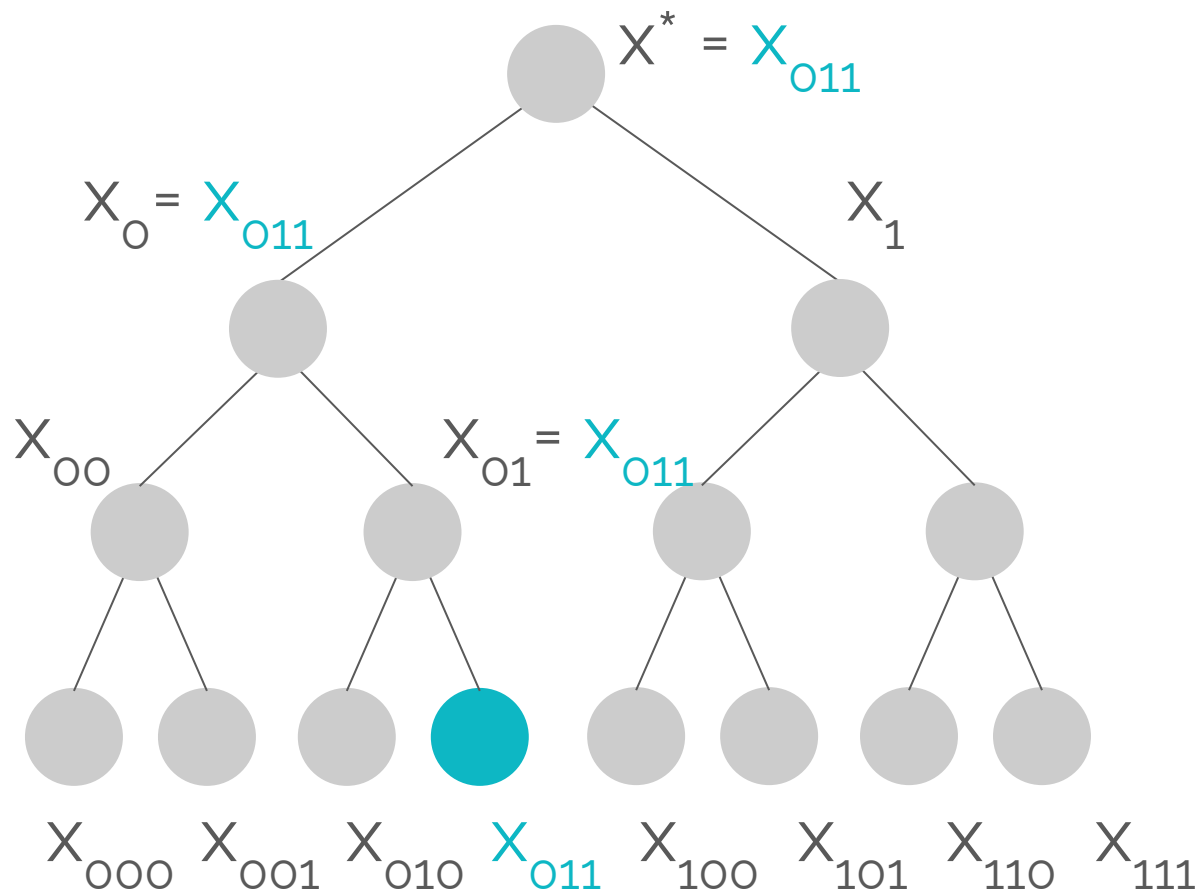
Avoid BARGs

through

non-blackbox

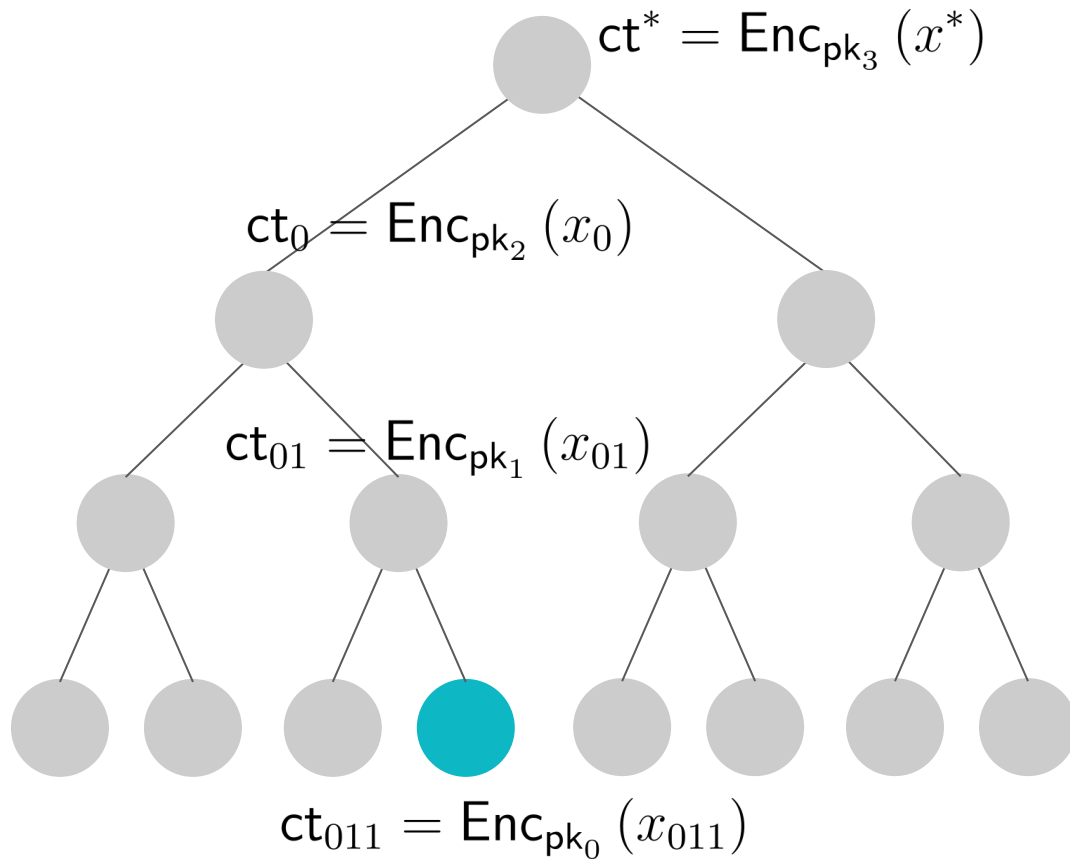
use of **SSB hash**

[HW15]



SSB hash

[HW15]



Public key

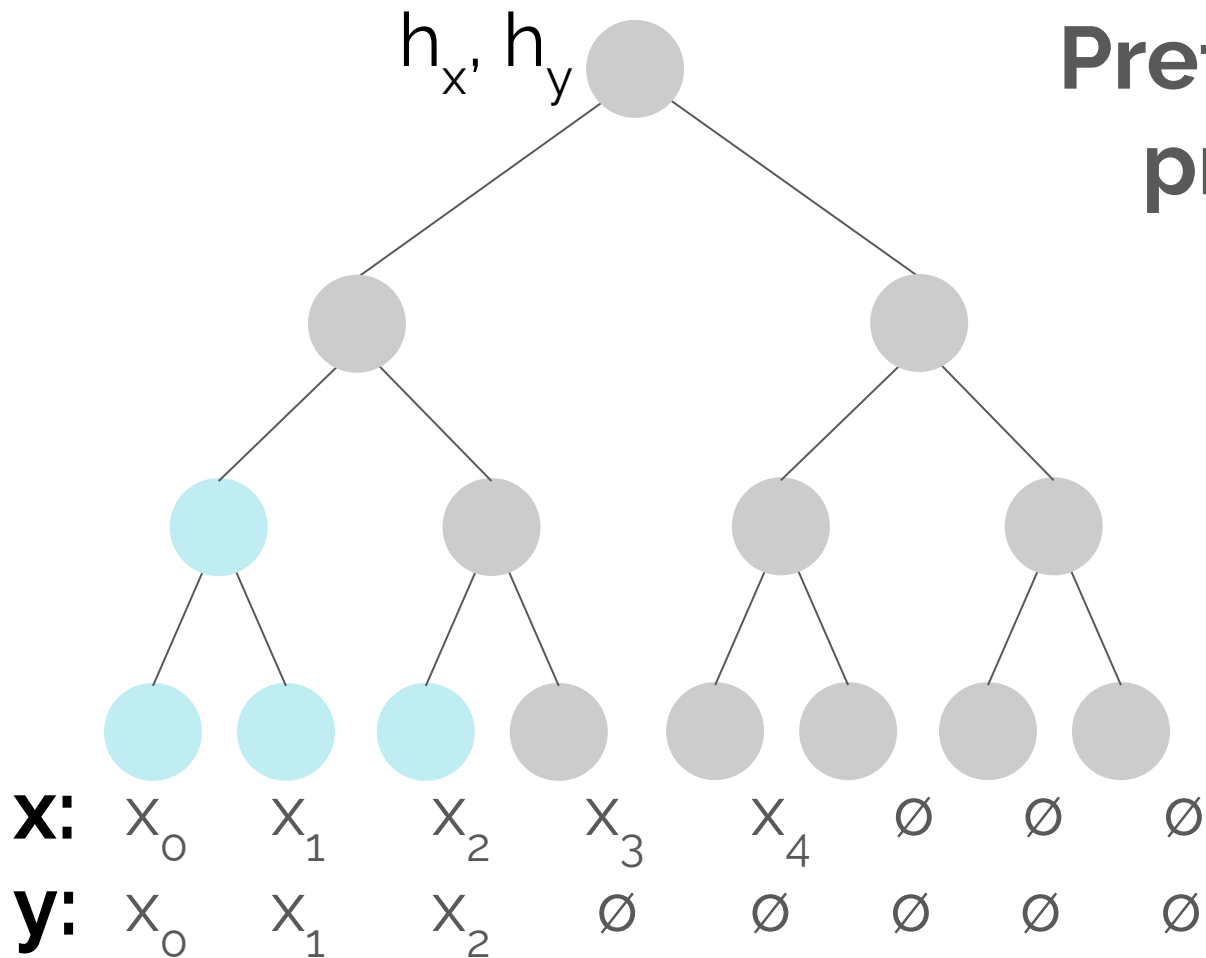
$pk_3, \text{Enc}_{pk_3}(sk_2, idx_3)$

$pk_2, \text{Enc}_{pk_2}(sk_1, idx_2)$

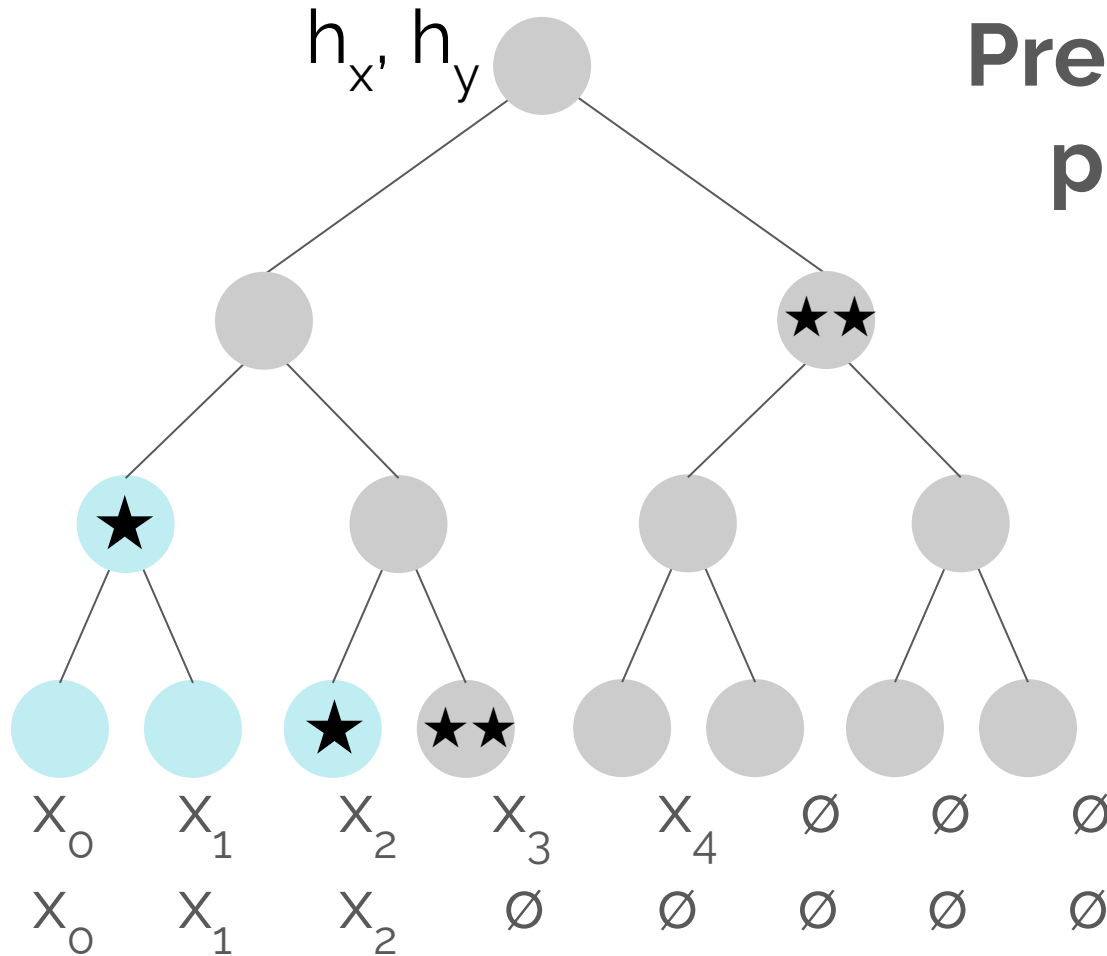
$pk_1, \text{Enc}_{pk_1}(sk_0, idx_1)$

pk_0

Prefix consistency proof for h_x, h_y

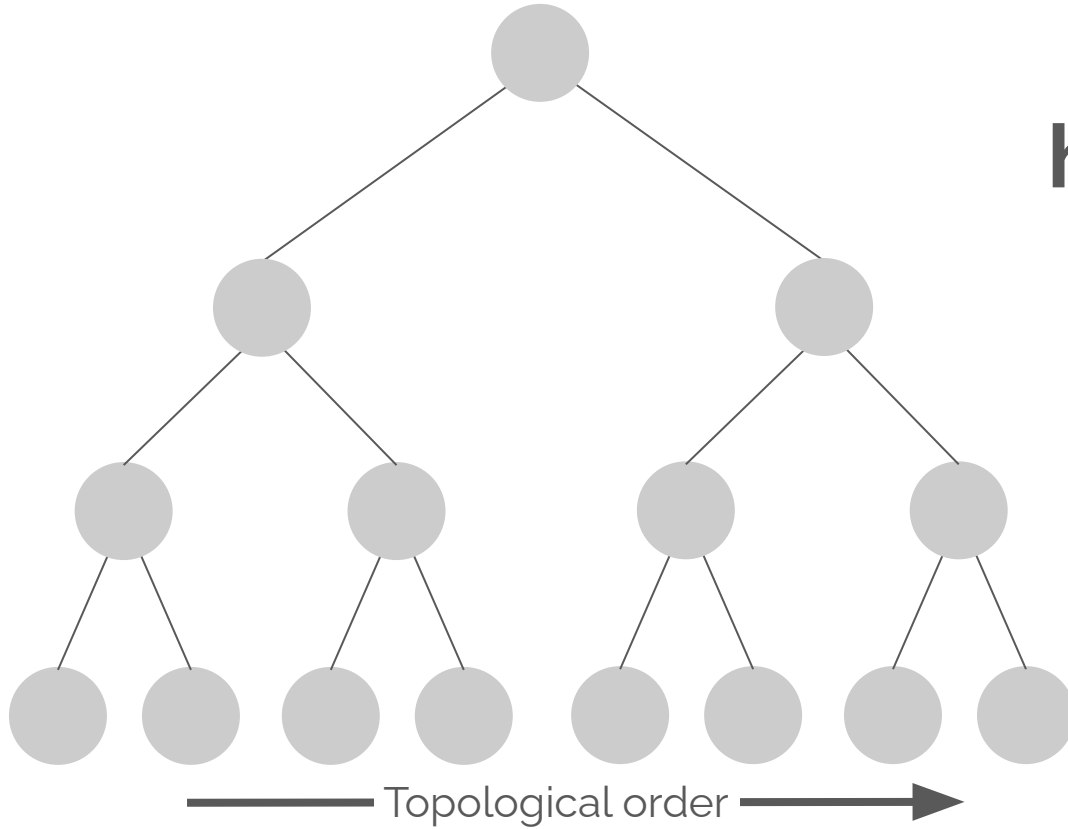


Prefix consistency proof for h_x, h_y



★: 1 hash
★★: 2 hashes

**Compute all N
hashes & proofs
incrementally
in $\sim N$ time**



every wire: hash dependent wires

every gate: prove input hash consistent with output hash

Also in our paper

[eprint/2025/307](#)

- Detailed construction and proofs
- Applications to MIFE
- Applications to iO for TM

Open questions

- **Prove/disprove the input-len barrier**
- **Concretely efficient iO?**

Thank you !

runting@gmail.com